DAY 13 – TCP AND UDP COMPARISON

Cyber Security Internship – Veda Technology

1. INTRODUCTION

TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) are two important transport-layer protocols used in computer networking.

Both protocols operate at Layer 4 of the OSI model and are responsible for communication between applications over a network.

TCP focuses on reliable and ordered data transmission, whereas UDP focuses on faster communication with minimal overhead.

Understanding the differences between TCP and UDP is important in cybersecurity, network troubleshooting, and selecting appropriate protocols for different applications.

2. OBJECTIVE

The main objective of this task is to compare TCP and UDP communication and identify their appropriate use cases.

Specific Objectives:

1. Understand TCP and UDP protocols.
2. Compare their reliability and connection behavior.
3. Identify differences in communication overhead.
4. Understand practical applications of TCP and UDP.
5. Learn their importance in network security.

3. TRANSMISSION CONTROL PROTOCOL (TCP)

TCP is a connection-oriented transport-layer protocol that provides reliable and ordered data delivery between applications.

Before transmitting application data, TCP establishes a connection using a three-way handshake.

Features of TCP

- Connection-oriented communication.
- Reliable data transmission.
- Error detection and retransmission.
- Ordered delivery of data.

- Flow control and congestion control.
- Higher protocol overhead compared to UDP.

TCP Three-Way Handshake

Step 1 – SYN: The client sends a SYN packet to request a connection.

Step 2 – SYN-ACK: The server responds with a SYN-ACK packet.

Step 3 – ACK: The client sends an acknowledgment to establish the connection.

Connection Establishment:

Client → SYN → Server

Client ← SYN-ACK ← Server

Client → ACK → Server

After completing the handshake, application data transmission can begin.

Applications of TCP

1. Web browsing using HTTP/1.1 and HTTP/2.
2. Email transmission using SMTP.
3. File transfer using FTP.
4. Remote access using SSH.
5. Secure communication using HTTPS over TCP.

4. USER DATAGRAM PROTOCOL (UDP)

UDP is a connectionless transport-layer protocol that allows applications to send data without establishing a connection.

UDP does not guarantee packet delivery, ordering, or retransmission.

It has lower protocol overhead and is useful for applications requiring low latency.

Features of UDP

- Connectionless communication.
- No three-way handshake.
- Low protocol overhead.
- No built-in retransmission.
- No guaranteed packet ordering.

- Suitable for real-time applications.

UDP Communication Process

1. The sender prepares a datagram.
2. The datagram is transmitted to the receiver.
3. No connection establishment is required.
4. The receiver processes the datagram if it arrives.
5. UDP itself does not provide delivery acknowledgments.

Applications of UDP

1. Online multiplayer gaming.
2. Voice over Internet Protocol (VoIP).
3. Live video streaming.
4. DNS queries.
5. DHCP communication.

5. TCP VS UDP COMPARISON TABLE

Feature	**TCP**	**UDP**
Full Form	Transmission Control Protocol	User Datagram Protocol
OSI Layer	Transport Layer	Transport Layer
Connection	Connection-oriented	Connectionless
Reliability	Reliable delivery	Best-effort delivery
Speed	Generally more overhead	Generally lower overhead
Handshake	Required	Not required
Packet Ordering	Guaranteed	Not guaranteed
Error Detection	Supported	Supported
Error Recovery	Retransmission	No built-in retransmission
Flow Control	Available	Not provided
Congestion Control	Built-in	Not built-in
Header Size	Minimum 20 bytes	8 bytes

Data Transmission	Ordered byte stream	Individual datagrams
Common Uses	Web, email, file transfer	Gaming, VoIP, DNS

6. RELIABILITY, CONNECTION BEHAVIOR AND OVERHEAD

6.1 Reliability

TCP:

- Uses acknowledgment mechanisms.
- Retransmits lost data.
- Ensures ordered delivery.
- Suitable for applications requiring accuracy and completeness.

UDP:

- Does not guarantee successful delivery.
- Does not retransmit lost packets automatically.
- Does not guarantee packet order.
- Suitable when applications prioritize low latency.

6.2 Connection Behavior

TCP:

TCP establishes a connection before transferring application data using the three-way handshake.

UDP:

UDP sends datagrams without establishing a transport-layer connection.

6.3 Communication Overhead

TCP:

- Minimum header size of 20 bytes.
- Requires connection establishment.
- Uses acknowledgments and retransmissions.
- Maintains connection state.

UDP:

- Fixed header size of 8 bytes.

- No connection-establishment handshake.
- No built-in acknowledgments or retransmissions.
- Minimal transport-layer overhead.

7. PRACTICAL USE CASES

Use Case 1 – File Download

Preferred Protocol: TCP

TCP is suitable for file downloads because the complete file must arrive correctly and in order.

Use Case 2 – Online Gaming

Preferred Protocol: UDP

UDP is commonly preferred for real-time game updates because low latency is important. Games can implement reliability for selected messages when needed.

Use Case 3 – Email Communication

Preferred Protocol: TCP

Email protocols such as SMTP use TCP to provide reliable data transmission.

Use Case 4 – Video Calling

Preferred Protocol: UDP

UDP is commonly used for real-time audio and video because delays can affect communication quality.

Use Case 5 – DNS Queries

Common Protocol: UDP

DNS commonly uses UDP for standard queries because of its low overhead. TCP is also supported and used when necessary.

8. IMPORTANCE OF TCP AND UDP IN CYBERSECURITY

TCP and UDP play important roles in network security and traffic analysis.

TCP Security Threats

- **SYN Flood Attack:** Attackers send large numbers of connection requests to exhaust server resources.
- **TCP Session Hijacking:** Attackers attempt to take control of an existing TCP communication session.

- **Port Scanning:** Attackers probe TCP ports to identify accessible services.

UDP Security Threats

- **UDP Flood Attack:** Attackers send excessive UDP traffic to overwhelm network resources.
- **UDP Amplification Attack:** Attackers exploit certain UDP-based services to generate large responses toward a victim.
- **IP Spoofing:** Attackers may forge source IP addresses in UDP packets.

Security Measures

1. Configure firewalls to filter unwanted traffic.
2. Monitor suspicious TCP and UDP activity.
3. Use intrusion detection and prevention systems.
4. Apply rate limiting to reduce flooding attacks.
5. Keep network services updated and properly configured.

9. TCP AND UDP PACKET ANALYSIS USING WIRESHARK

Wireshark is a network protocol analyzer that can be used to observe TCP and UDP communication.

TCP Packet Analysis

1. Open Wireshark.
2. Select the active network interface.
3. Start packet capture.
4. Generate TCP traffic by accessing a suitable website or TCP-based service.
5. Apply the display filter tcp.
6. Observe source and destination ports, sequence numbers, acknowledgments, and TCP flags.

Expected Observation:

TCP traffic may show SYN, SYN-ACK, ACK, and other packets used for connection management and reliable data delivery.

UDP Packet Analysis

1. Start packet capture in Wireshark.

2. Generate UDP traffic using a DNS lookup or another UDP-based application.
3. Apply the display filter udp.
4. Observe source and destination ports, datagram length, and checksum fields.

Expected Observation:

UDP traffic does not contain TCP-style connection establishment or transport-layer acknowledgment mechanisms.

Note: These are expected observations. Actual results depend on the captured network traffic.

10. CONCLUSION

TCP and UDP are essential transport-layer protocols used in computer networking.

TCP provides reliable, connection-oriented communication, making it suitable for file transfers, email, and applications requiring ordered data delivery.

UDP provides connectionless communication with lower overhead, making it suitable for real-time applications such as gaming, voice calls, and live streaming.

Through this task, I understood the differences between TCP and UDP in terms of reliability, connection behavior, protocol overhead, and practical applications.

This knowledge is important for cybersecurity because it helps in network traffic analysis, identifying protocol-based threats, and implementing appropriate security measures.